

Question 1 : What is Simple Linear Regression (SLR)? Explain its purpose.

Answer:

Simple Linear Regression (SLR) is a **statistical method** used to study the relationship between **two continuous variables** — one **independent variable (predictor)** and one **dependent variable (response)**.

It models this relationship by fitting a **straight line** (called the *regression line*) to the observed data, expressed by the equation:

$$Y = \beta_0 + \beta_1 X + \epsilon$$

Where:

- Y = Dependent variable (the outcome we want to predict)
- X = Independent variable (the predictor)
- β_0 = Intercept (value of Y when X = 0)
- β_1 = Slope (how much Y changes for a unit change in X)
- ϵ = Error term (difference between predicted and actual values)

Purpose of Simple Linear Regression:

- 1. Prediction:**
To predict the value of the dependent variable based on the known value of the independent variable.
Example: Predicting sales (Y) based on advertising expenditure (X).
- 2. Understanding Relationships:**
To determine the **strength and direction** of the relationship between the two variables (positive or negative correlation).
- 3. Trend Estimation:**
To identify general trends or patterns in data — for instance, whether sales increase with price reduction.
- 4. Decision Making:**
It helps businesses, researchers, and analysts make data-driven decisions using predictive insights.

Question 2: What are the key assumptions of Simple Linear Regression?

Answer:

Simple Linear Regression (SLR) relies on several key **assumptions** to ensure that the model's estimates and predictions are accurate and reliable. These assumptions must be checked before interpreting the results.

Key Assumptions of Simple Linear Regression:

1. Linearity:

- The relationship between the independent variable (**X**) and the dependent variable (**Y**) is **linear**.
- This means that a change in X leads to a proportional change in Y.
- *Example:* If advertising spending doubles, sales roughly double (linear pattern).

2. Independence of Errors:

- The residuals (errors) are **independent** of each other.
- There should be no correlation between error terms from different observations.
- *Violation Example:* In time series data, today's error might depend on yesterday's error (autocorrelation).

3. Homoscedasticity (Constant Variance of Errors):

- The variance of residuals (errors) is **constant** across all levels of the independent variable.
- *Violation Example:* If the spread of residuals increases with X, it indicates **heteroscedasticity**.

4. Normality of Errors:

- The residuals (errors) should be **normally distributed**.

- This assumption is crucial for conducting hypothesis tests and building confidence intervals.
- *Example:* A histogram or Q–Q plot of residuals should show a normal distribution shape.

5. No Multicollinearity:

- Although SLR involves only one independent variable, this assumption becomes critical in **multiple regression**.
- In SLR, it simply means the independent variable should not be constant or have very little variation.

6. No Measurement Error in the Independent Variable:

- The independent variable (X) is assumed to be measured **without error**.
- Any measurement error in X can lead to biased slope estimates.

Question 3: Write the mathematical equation for a simple linear regression model and explain each term.

Answer:

The mathematical equation for a Simple Linear Regression (SLR) model is:

$$Y = \beta_0 + \beta_1 X + \epsilon$$

Explanation of Each Term:

1. Y – Dependent Variable (Response Variable):

- This is the variable we are trying to predict or explain.
- Example: Sales, marks, house price, etc.

2. X – Independent Variable (Predictor Variable):

- This is the variable that is used to predict the value of Y .
- Example: Advertising expenditure, study hours, house size, etc

3. β_0 – Intercept (Constant Term):

- The value of Y when $X = 0$.
- It represents the point where the regression line crosses the Y -axis.
- Example: If $\beta_0 = 50$, then even when advertising = 0, sales are expected to be 50 units.

4. β_1 – Slope (Regression Coefficient):

- It indicates the change in Y for a one-unit change in X .
- If $\beta_1 = 2$, it means that for every 1 unit increase in X , Y increases by 2 units (assuming other factors remain constant).

5. ϵ – Error Term (Residual):

- Represents the difference between the observed value and the predicted value.
- It accounts for all the factors that affect Y but are **not included** in the model.
- Mathematically: $\epsilon = Y_{\text{actual}} - Y_{\text{predicted}}$.

Question 4: Provide a real-world example where simple linear regression can be applied.

Answer:

A real-world example of applying Simple Linear Regression (SLR) is in predicting house prices based on house size (area in square feet).



Example: Predicting House Prices

- Objective:
To predict the *price of a house* (dependent variable) based on its *size in square feet*

(independent variable).

Model Equation:

$$\text{Price} = \beta_0 + \beta_1(\text{Size}) + \epsilon$$

Where:

- **Price (Y):** The dependent variable — the price of the house (in ₹ or \$).
- **Size (X):** The independent variable — the area of the house in square feet.
- **β_0 :** The intercept — the base price when house size = 0 (theoretical value).
- **β_1 :** The slope — how much the house price increases per additional square foot.
- **ϵ :** The error term — factors not included in the model (e.g., location, condition, age of house).

Other Real-Life Examples:

1. Predicting **sales** based on **advertising spend**.
2. Estimating **student marks** based on **study hours**.
3. Forecasting **fuel consumption** based on **vehicle speed**.
4. Predicting **crop yield** based on **rainfall amount**.

Question 5: What is the method of least squares in linear regression?

Answer:

The Method of Least Squares is the most common technique used to find the best-fitting line in Simple Linear Regression.

Its goal is to minimize the sum of the squared differences between the actual values and the predicted values from the regression line.

Concept:

In Simple Linear Regression, the model is:

$$Y = \beta_0 + \beta_1 X + \epsilon$$

Here, ϵ represents the error (residual) — the difference between the actual value (Y_i) and the predicted value ($\hat{Y}_i$):

$$\epsilon_i = Y_i - \hat{Y}_i$$

Least Squares Principle:

The **method of least squares** finds the line that **minimizes the sum of squared residuals**, i.e.:

$$\text{Minimize } S = \sum_{i=1}^n (Y_i - \hat{Y}_i)^2 = \sum_{i=1}^n (Y_i - (\beta_0 + \beta_1 X_i))^2$$

The values of β_0 and β_1 are chosen such that this **sum of squared errors (S)** is as small as possible.

Interpretation:

- The least squares line is the one that makes the **total squared prediction error** as small as possible.
- It ensures that the line passes **as close as possible to all data points** in terms of squared vertical distances.

Question 6: What is Logistic Regression? How does it differ from Linear Regression?

Answer:

Feature	Linear Regression	Logistic Regression
Purpose	Predicts continuous outcomes	Predicts categorical outcomes (usually binary)
Output Range	Any real number ($-\infty$ to $+\infty$)	Between 0 and 1 (probability)
Equation Type	Straight line: $(Y = \beta_0 + \beta_1 X)$	Sigmoid curve: $(P(Y=1) = 1 / (1 + e^{-(\beta_0 + \beta_1 X)}))$

Error Function	Minimizes sum of squared errors	Uses maximum likelihood estimation
Assumptions	Assumes linear relationship between X and Y	Assumes linear relationship between X and the log-odds of Y
Use Cases	Predicting house prices, sales, income, etc.	Predicting pass/fail, disease/no disease, spam/not spam, etc.

- **Linear Regression** → predicts **continuous values** using a straight line.
- **Logistic Regression** → predicts **probabilities (0–1)** for **categorical outcomes** using a **sigmoid (S-shaped) curve**.

Question 7: Name and briefly describe three common evaluation metrics for regression models.

ANSWER:

There are several metrics used to evaluate how well a **regression model** fits the data. The three most common evaluation metrics are:

1. Mean Absolute Error (MAE):

$$MAE = \frac{1}{n} \sum_{i=1}^n |Y_i - \hat{Y}_i|$$

Meaning:

It measures the **average absolute difference** between the actual values (Y_i) and predicted values ($\hat{Y}_i$).

- **Interpretation:**
Lower MAE → better model performance.
- **Advantage:**
Easy to understand since it uses the same units as the data.

- **Example:**
If MAE = 5, the model's predictions are off by 5 units on average.
-

2. Mean Squared Error (MSE):

$$MSE = \frac{1}{n} \sum_{i=1}^n (Y_i - \hat{Y}_i)^2$$

- **Meaning:**
It measures the **average of the squared differences** between actual and predicted values.
 - **Interpretation:**
Smaller MSE indicates a better fit.
 - **Advantage:**
Penalizes large errors more strongly (since errors are squared).
 - **Drawback:**
Not in the same units as the target variable (because of squaring).
-

3. R-squared (Coefficient of Determination):

$$R^2 = 1 - \frac{\sum (Y_i - \hat{Y}_i)^2}{\sum (Y_i - \bar{Y})^2}$$

Meaning:

It represents the **proportion of variance** in the dependent variable that is explained by the model.

- **Range:**
 $0 \leq R^2 \leq 1$
- **Interpretation:**
 - $R^2 = 1$: Perfect fit (model explains all variance).
 - $R^2 = 0$: Model explains no variance (poor fit).
- **Example:**
 $R^2 = 0.85 \rightarrow$ The model explains **85%** of the variation in the dependent variable.

Question 8: What is the purpose of the R-squared metric in regression analysis?

Answer:

Purpose of the R-squared Metric in Regression Analysis:

The **R-squared (R^2)** metric, also known as the **Coefficient of Determination**, is used to measure how **well a regression model explains the variability** in the dependent variable using the independent variable(s).

Mathematical Formula:

$$R^2 = 1 - [SS_{res} / SS_{tot}]$$

Where:

- $SS_{res} = \sum (Y_i - \hat{Y}_i)^2 \rightarrow$ **Residual Sum of Squares** (unexplained variation)
- $SS_{tot} = \sum (Y_i - \bar{Y})^2 \rightarrow$ **Total Sum of Squares** (total variation in data)

Interpretation:

- R^2 represents the **proportion of variance in the dependent variable (Y) that is explained by the regression model**.
- It tells us **how well the model fits the data**.

R^2 Value	Interpretation
0	Model explains none of the variability in Y
0.5	Model explains 50% of the variability in Y
1	Model perfectly explains all variability in Y

xample:

If $R^2 = 0.85$

→ 85% of the variation in the dependent variable (e.g., sales) is explained by the independent variable (e.g., advertising spend).

→ The remaining 15% is due to other factors or random noise not captured by the model.

Purpose / Importance:

1. **Model Fit Indicator:** Shows how well the model describes the data.
2. **Comparison Tool:** Helps compare different regression models — higher R^2 generally indicates a better fit.

Question 9: Write Python code to fit a simple linear regression model using scikit-learn and print the slope and intercept.

(Include your Python code and output in the code box below.)

```
1 # Import necessary libraries
2 from sklearn.linear_model import LinearRegression
3 import numpy as np
4
5 # Example data
6 # X = independent variable (e.g., hours studied)
7 # y = dependent variable (e.g., marks obtained)
8 X = np.array([2, 4, 6, 8, 10]).reshape(-1, 1)
9 y = np.array([81, 93, 91, 97, 109])
10
11 # Create a Linear Regression model
12 model = LinearRegression()
13
14 # Fit the model to the data
15 model.fit(X, y)
16
17 # Get the slope (coefficient) and intercept
18 slope = model.coef_[0]
19 intercept = model.intercept_
20
21 # Print the results
22 print("Slope ( $\beta_1$ ):", slope)
23 print("Intercept ( $\beta_0$ ):", intercept)
24
```

```
➦ Slope ( $\beta_1$ ): 3.0
  Intercept ( $\beta_0$ ): 76.2
```

Question 10: How do you interpret the coefficients in a simple linear regression model?

Answer:

In a **Simple Linear Regression** model, the equation is:

$$Y = \beta_0 + \beta_1 X + \varepsilon$$

Where:

- β_0 = Intercept
- β_1 = Slope (Coefficient of X)
- ϵ = Error term

1. Intercept (β_0):

- The **intercept** represents the **predicted value of Y when X = 0**.
- It indicates the starting point or baseline of the dependent variable.
- **Example:**
If $\beta_0=50$, it means that when X (e.g., study hours) = 0, the predicted Y (e.g., marks) is 50.

Note: The intercept may not always have a practical meaning if X = 0 is unrealistic (e.g., house size = 0 sq.ft).

2. Slope (β_1):

- The **slope coefficient** represents the **change in the dependent variable (Y) for a one-unit increase in the independent variable (X)**.
- It tells us the **direction** and **strength** of the relationship:
 - If $\beta_1 > 0$: Positive relationship — Y increases as X increases.
 - If $\beta_1 < 0$: Negative relationship — Y decreases as X increases.
- **Example:**
If $\beta_1=3.2$, it means that for every **1 unit increase in X**, Y increases by **3.2 units**, on average.

Example Interpretation:

Suppose the regression equation is:

$$\text{Sales} = 50 + 2.5 \times \text{Advertising}$$

- **Intercept ($\beta_0 = 50$):** When no money is spent on advertising, expected sales = 50 units.
- **Slope ($\beta_1 = 2.5$):** For every additional ₹1 spent on advertising, sales increase by 2.5 units.